# Rhizome Testing Day

A manual QA strategy for the rhizome window manager

River + StumpWM-parity stack — Sunday edition

## 1 Goal and method

Rhizome now has broad feature coverage; this plan puts it through its paces as a **daily driver** under real usage, not just unit conditions. The strategy has three layers:

1. **Regression sweep** — run the automated suite first, so manual testing starts from a known-good base.
2. **Feature circuits** — scripted walks through each feature area with concrete steps and expected results (the bulk of this document).
3. **Soak testing** — use rhizome as your only window manager for the day: browse, edit, chat, take screenshots, switch workspaces under load, and note anything that breaks, flickers, or feels wrong.

Mark each item with a result. Anything that fails: note the section number, what you did, what happened — the capture guide in the appendix turns that into a fixable report.

### 1.1 Preconditions

☐ Install missing helpers: `sudo pacman -S swaylock swayidle` (lock screen — MOD+Shift+L is a no-op without them). ☐ Fresh login: pick the rhizome (river) session at the SDDM greeter. Watch the first 10 seconds: one bar, one wallpaper process, both autostart apps (variety tray-less in tray, flameshot daemon). ☐ Automated suite green (next section).

### 1.2 The automated suite (15 min)

Run each from the repo root; every line must end in `0 failed`:

```
for t in workspace-restore workspace-switch-away window-rules \
         hooks-and-eval ipc prefix prompt-and-message floating \
         frames parity quickwins interactivity medium3 bigger1 \
         session-icons bar-guard expose; do
  echo "== $t"; sbcl --non-interactive --load tests/$t.lisp
done
```

| Suite | What it covers | Result |
|---|---|---|
| frames (57) | Tree ops, dynamic tiling, dumps, refits | ☐ |
| expose (22) | Grid, pick paths, exact restore | ☐ |
| scroll (27) | Ribbon geometry, clamping, focus recovery | ☐ |
| groups (11) | Group types, floating adoption, cycling | ☐ |
| dbus (14) | Wire protocol + live broker integration | ☐ |
| tray (13) | Watcher service, item registry, click routing | ☐ |
| bigger1 (27) | Minor modes, WSE, fuzzy | ☐ |
| session-icons (25) | Lock/logout commands, bar icons | ☐ |
| the rest (152) | Everything else, per README | ☐ |

Total: 414 checks. If any fail, stop — manual testing is pointless on a broken base.

# 2 Feature circuits

## 2.1 B. Dynamic tiling (the new core)

The behavior under the most scrutiny today: every new window should get its own frame.

| | Action | Expected |
|---|---|---|
| ☐ | On an empty workspace press MOD+Return twice | First terminal fills the screen; the second splits it side-by-side |
| ☐ | MOD+Return again (3rd, 4th, 5th window) | Each arrival splits the frame you are in along its longer axis; the pointer follows the newest window |
| ☐ | C-t R (remove split) after several arrivals | Frames merge pairwise into their sibling; windows stack one-per-frame again |
| ☐ | Close windows one by one with several open | The empty frame is reclaimed: neighbors grow automatically — no manual C-t R needed |
| ☐ | C-t Q (only) with several frames | Whole workspace collapses to one frame; all windows in it, cycle with C-t n |
| ☐ | C-t B (balance) after manual resizing | Every split returns to 50/50 |
| ☐ | Open 8+ windows | Splits stop below twice the minimum frame size (no sliver frames); still stable |
| ☐ | C-t n / C-t p inside a multi-window frame | Pulls the next/previous hidden window into the current frame |

## 2.2 C. Frame tree surgery

| | Action | Expected |
|---|---|---|
| ☐ | C-t s, C-t S | Manual side-by-side and stacked splits; focus moves to the new empty frame |
| ☐ | C-t arrows | Focus moves to the nearest frame in that direction, preferring edge-sharing neighbors |
| ☐ | C-t +/- | Dividing line nudges ±30px; iresize (C-t r) then arrows works continuously, Enter commits, Escape cancels |
| ☐ | C-t M + arrows | Moves the focused window into that frame |
| ☐ | C-t e + arrows | Swaps the current frame's window with the target frame's (focus follows the arrival) |
| ☐ | Switch layout: `rhizome-cmd switch-layout master-stack` | Tree rebuilds as master/stack; windows distributed one per frame |
| ☐ | `rhizome-cmd dump-frame proj` then open/close windows, then `restore-frame proj` | Layout returns; windows placed by number; missing windows skipped without crash |

## 2.3 D. Expose (C-t Return)

| | Action | Expected |
|---|---|---|
| ☐ | With 3+ windows on the workspace: C-t Return | All windows (including hidden-in-frame ones) float into an aspect-fitted grid; a hint message appears |

| | | |
|---|---|---|
| ☐ | Press 1–9 / 0 | That grid slot is picked; layout restores exactly and that window focuses |
| ☐ | Arrows, then Return | Focus walks the grid; Return picks the focused window |
| ☐ | Click a window in the grid | Pick completes with that window |
| ☐ | Escape | Layout and focus restore to pre-expose state exactly — float positions, frame order, focus |
| ☐ | Wait 60s mid-expose | Timeout restores safely on its own |
| ☐ | One window only: C-t Return | “nothing to expose” message, no state change |

### 2.4 D2. The scroll ribbon (C-t v, niri-style)

| | Action | Expected |
|---|---|---|
| ☐ | C-t v with 2-3 windows open | Windows become full-height columns in their tiling order; the bar’s layout module reads “scroll”; a message reports the column count |
| ☐ | C-t . / C-t , (period/comma) | Focus moves one column over; the viewport scrolls the minimum distance to keep the focused column fully visible |
| ☐ | MOD+. / MOD+, | Viewport scrolls alone in half-screen steps, clamped at both ribbon ends; focus unchanged; the scroll-x message appears |
| ☐ | Open a new window while scrolled | It enters as a new column immediately right of the focused one and gets focus |
| ☐ | Close the focused window while scrolled | Its column is pruned; focus falls to the vacated slot’s successor, not the first column |
| ☐ | Float a column window (C-t f) | It leaves the ribbon (column count drops); the float renders and drags normally |
| ☐ | C-t v again | Back to the dynamic tiler; windows re-split into frames as if the ribbon never happened |
| ☐ | C-t Return (expose) while scrolled | The overview works over columns; restore returns to the ribbon exactly |

### 2.5 D2b. The tray (StatusNotifier icons)

| | Action | Expected |
|---|---|---|
| ☐ | Bar with Variety running | A tray icon appears in the right segment next to the clock; the bar log shows the item registered |
| ☐ | Left-click a tray icon | The item’s Activate runs (Variety opens its preferences, flameshot its overlay) |
| ☐ | Scroll over a tray icon | Scroll(delta, vertical) reaches the item without errors in the log |

| | | |
|---|---|---|
| ☐ | Close an app with a tray item | Its icon disappears within a second (NameOwnerChanged fires and the item is pruned) |
| ☐ | `busctl --user status org.kde.StatusNotifierWatcher` | PID points at the bar's sbcl process |
| ☐ | `rhizome-cmd toggle-bar` twice | The watcher name follows the bar (released on stop, re-acquired on start); icons return |

## 2.6 D2c. The display configurator (bar :display module)

| | Action | Expected |
|---|---|---|
| ☐ | Bar right segment | Monitor icon + brightness % (e.g. "100") after battery |
| ☐ | Click the display module | Display menu opens: brightness steps, night light, "Output eDP-2 (WxH @ RHz)" |
| ☐ | Brightness +10% | Brightness steps up; the bar readout follows within the interval |
| ☐ | Toggle night light | hyprsunset starts (warm tint); toggle again kills it |
| ☐ | Output entry → submenu | Mode list with current marked ✓; scale options 1/1.25/1.5/2 |
| ☐ | Pick a non-current mode | "display: applied" message; resolution changes; pick the old mode to restore |
| ☐ | Pick a scale | Scale changes; pick the original to restore |
| ☐ | `rhizome-cmd "(ql:quickload :rhizome)"` then click display | Output entries still present (defvar state survives reloads) |

## 2.7 D3. Group types (C-t l)

| | Action | Expected |
|---|---|---|
| ☐ | C-t l with tiled windows | The ribbon appears (tile → scroll); message names the type |
| ☐ | C-t l again | Everything floats in a cascade (scroll → floating); message names the type |
| ☐ | Open a new window while floating | It arrives floated, cascaded below the existing floats — not tiled |
| ☐ | C-t l a third time | Back to dynamic tiling (floating → tile); windows re-split into frames |
| ☐ | `rhizome-cmd set-group-type floating` then back with `tile` | Direct type sets behave like the cycle legs |

## 2.8 E. Floats and pointer operations

| | Action | Expected |
|---|---|---|
| ☐ | C-t f on a tiled window | Floats in place (no jump); "floated" message |
| ☐ | MOD+Left-drag a float | Moves; alpha-blended border follows |
| ☐ | MOD+Left-drag a **tiled** window | StumpWM-style: it floats first, then moves |

| | | |
|---|---|---|
| ☐ | MOD+Right-drag a float | Resizes from the bottom-right; respects min/max size hints |
| ☐ | Drag a float mostly off-screen, release | Clamped back so  30px stays reachable |
| ☐ | Titlebar-drag a CSD app (GIMP toolbox, if installed) | Moves without floating it again |
| ☐ | C-t f again to unfloat | Returns to tiling in the current frame |
| ☐ | Drag a float across a workspace switch (MOD+1..9 while dragging is fine to skip) | No crash; float lands on its workspace |

### 2.9 F. Workspaces, rules, marks

| | Action | Expected |
|---|---|---|
| ☐ | MOD+1..9 / C-t 1..9 | Direct switch; bar highlights; hidden windows restore |
| ☐ | MOD+Shift+1..9 | Moves the focused window and follows it |
| ☐ | C-t N / C-t P | Next/previous workspace |
| ☐ | `rhizome-cmd renumber` after closing windows | Numbers compact 1..N; C-t ' selects by number |
| ☐ | C-t # then `rhizome-cmd move-marked 3` | Marked window (star) moves; marks clear |
| ☐ | Set a rule: `(configure '(:window-rules ((:match "foot" :workspace 4))))`, then launch a terminal | It opens on workspace 4; check via C-t w (windowlist shows workspace numbers) |
| ☐ | Same with a `:wse` rule: `(:wse (:and (:app-id "foot") (:float nil)) :float t)` | Matching windows float automatically |

### 2.10 G. Prompts, launcher, menus (fuzzy)

| | Action | Expected |
|---|---|---|
| ☐ | MOD+D, type “ff” | “Firefox”-style fuzzy hits rank after exact substring matches; the just-installed Variety appears |
| ☐ | MOD+W (windowlist) | Numbers shown as `N: [ws] title`; marked windows show a star; Enter jumps to its workspace and focuses |
| ☐ | MOD+X (command palette), type “bal” | Fuzzy filters; Enter runs balance; the message window confirms |
| ☐ | C-t ' then a number | Selects that window by number |
| ☐ | Tab/Up/Down in any prompt | Selection cycles; Escape cancels cleanly every time |
| ☐ | Leave a prompt open 30s | Auto-closes; keyboard never trapped |

### 2.11 H. Bar

| | Action | Expected |
|---|---|---|
| ☐ | Click workspace numbers in the bar | Switches workspaces via IPC |
| ☐ | Click the flameshot icon | Opens flameshot gui |
| ☐ | Click the variety icon | Launches/raises Variety preferences |
| ☐ | Scroll on the volume module | Volume steps 5% |

| | | |
|---|---|---|
| ☐ | C-t b / MOD+B | Bar hides and returns; frame trees refit — windows never tiled under or floating over the bar |
| ☐ | `(configure '(:bar (:height 30)))` | Bar restarts taller; windows re-tile under it immediately |

### 2.12 I. Session management

| | Action | Expected |
|---|---|---|
| ☐ | MOD+Shift+L | Locks (needs swaylock installed); password unlocks |
| ☐ | MOD+Shift+E once | “press again to log out” message |
| ☐ | MOD+Shift+E twice within 3s | Session ends; SDDM greeter returns; log back in cleanly |
| ☐ | `rhizome-cmd suspend` | Machine suspends; wake restores session with input working |
| ☐ | MOD+Shift+Q / R | Confirm-style behavior consistent; power actions prompt polkit as configured |
| ☐ | C-t q twice / once | Quit needs the second press; a single press just warns |

### 2.13 J. Prefix chains, help, modes

| | Action | Expected |
|---|---|---|
| ☐ | Press C-t and pause | Which-key list shows every binding with docs (multi-line message window) |
| ☐ | C-t h then descend b/c/f/v/k/w | Each shows its output; k then a key describes it without running it |
| ☐ | C-t C (command mode), run 3 commands, Escape | Stays armed between commands; which-key visible; Escape exits |
| ☐ | `(enable-minor-mode :test)` with a bound key | Mode binding shadows the prefix binding while enabled; disable restores |
| ☐ | C-t m (lastmsg), C-t a (time) | Both show in the message window; lastmsg re-shows the previous text |
| ☐ | Unbound key in a chain | “unbound key” message, clean exit |
| ☐ | `(set-prefix-key "C-z")`, try C-z then C-t | C-z now opens chains; C-t is inert; set it back with `"(rhizome:set-prefix-key \"C-t\")"` |
| ☐ | `(set-user-mod :alt)` and back to `:super` | MOD bindings re-grab in both directions; MOD+drag follows |

### 2.14 K. IPC and REPL robustness

| | Action | Expected |
|---|---|---|
| ☐ | `rhizome-cmd current-workspace`, `wse (:float t)`, `renumber` | All reply in the message window without breaking the WM |
| ☐ | `rhizome-cmd '(+ 1 2)'` | Evaluates; reply shows in the message window |

| | | |
|---|---|---|
| ☐ | M-x slime-connect (or `rhizome-cmd`) while windows are moving | No request from a foreign thread errors in the log |
| ☐ | Flood: hold MOD+J/K for 10s, then C-t chains immediately | No input loss, no stuck modes |
| ☐ | `rhizome-cmd lock` twice in a row | Second lock doesn't spawn duplicates (swaylock may refuse politely) |

## 3 Soak testing (the afternoon)

Use rhizome as your only session for 3–4 hours. Keep this list open and note anything that matches:

| Watch for | Notes |
|---|---|
| ☐ Flicker | Bar redraws, message window flashes, window borders re-drawn repeatedly — check the WM log for manage/render churn (a past bug class) |
| ☐ Focus fights | Auto-focus stealing, clicks not moving focus, expose leaving stale focus |
| ☐ Slow chains | C-t taking > 100ms, launcher scrolling hitching |
| ☐ Workspace drift | Windows landing on wrong workspaces via rules, or losing their frame order after switches |
| ☐ Log growth | Tail `~/.cache/rhizome/rhizome.log` and `bar.log` — idle should add nothing; bursts during drag/resize are expected but should stop on release |
| ☐ Memory | `ps -o rss -p $(pgrep -f rhizome:run)` at start vs end of the afternoon; >2x growth is worth investigating |

Suggested load: 8–12 windows across 4+ workspaces, a browser with video, an editor, two terminals, one GIMP-style float if available. Exercise expose, splits, and workspace switches while the video plays.

## 4 Advanced circuits (if the morning goes well)

- **Stability — protocol stream (the freeze fix):** open and close the launcher, show messages (C-t m), open prompts (MOD+W) repeatedly — 10+ popups. Then check the log: `grep -c "Malformed" ~/.cache/rhizome/rhizome.log` — reads 0 after the v5 compositor cap (any non-zero number means the desync is back; stop and report).
- **Stability — the snapshot storm (the freeze fix):** with the bar running and windows open, run `ls -l --time-style=+%s ~/.cache/rhizome/snapshot.sexp` twice, 10 seconds apart — the mtime must be IDENTICAL while nothing changes (a bumping mtime at idle = the bar feedback storm). Also: `tail -f ~/.cache/rhizome/rhizome.log` should be quiet at idle — a stream of MANAGE-START lines at idle is the storm.
- **Stability — scroll gauntlet:** toggle C-t v, wait 10s (the message expiry), toggle back, wait 10s. The laptop must stay responsive and the log quiet; this exact sequence froze the machine before commit b1e7996.
- **Session — portal dialogs (the chooser fix):** in Steam, open Settings → Storage → add a library folder — the omanta file picker must appear. This exercises D-Bus-activated portal backends, which died silently before the session imported its environment (river/init now mirrors Omarchy's dbus-update-activation-environment and /.config/xdg-desktop-portal/river-portals.conf pins the backends). Any app's file chooser counts: Firefox uploads, Chromium saves — if a picker never appears, the activation environment went stale again.
- **Session — systemd services (the anchor):** after login, `systemctl --user is-active graphical-session.target` must read `active`, and the session set must be running (`systemctl --user list-`

`units | grep -cE 'fcitx5|sleep-lock|crash-watch|owed|voxtype'` — at least 5). Before the rhizome-session.target anchor, none of these omarchy services ran in a river session at all. After a MOD+Shift+E logout and re-login, the services must come up exactly once — no duplicates.

- **Session — polkit prompt (the agent):** run `pkexec true` — the lxqt-policykit dialog must appear and, after entering the password, exit 0. Without a registered agent polkit falls back to a textual prompt that cannot render under Wayland, and pkexec fails with "Error creating textual authentication agent". This is also what the MOD+Shift+Q / R power rows exercise.
- **Hot-reload resilience:** `(rhizome:configure '(:bar (:height 26)))` from `rhizome-cmd`, then `ql:quickload` a changed file via the eval prompt (C-t :). The WM should survive; note any stuck mode or lost focus. Recovery lever: `C-t q q` then relogin.
- **Multi-bar safety:** confirm exactly one bar process at all times: `pgrep -fc "rhizome:run-bar"` must read `1` (the fork-bomb guard).
- **Timeout paths:** enter iresize, expose, command mode and just wait 60s in each — all three must restore their state cleanly on expiry.
- **Float + expose interplay:** a pre-existing float plus dynamic tiles, expose, pick, repeat — float geometry must survive every round trip.
- **Maximize round-trip:** on a tiled window run `rhizome-cmd maximize` — it must float covering the output minus the bar strip; maximize again must return it to the tiling. Repeat with a pre-existing float: maximize must save and restore its exact geometry. A client-initiated maximize (terminal set_maximized, browser maximize button) must land in the same state.
- **Theming — fonts and colors:** change `(rhizome:configure '(:theme (:font-size 16)))` live — borders and the in-WM surfaces (message window C-t m, prompt MOD+W, launcher MOD+D) must re-render with the larger font immediately; `toggle-bar` twice must restart the bar with the new font. Set `:theme :font-family "nonexistentxyz"` — fontconfig must fall back to a system font, never a crash or blank bar. Override `:launcher :background` and `:theme :focused-border` and confirm both surfaces picked them up on the next redraw.
- **Workspace indicator:** with windows on workspaces 1 and 2 only, the bar must show 1 and 2 in the foreground color, 3-9 muted, and the active workspace underlined in accent with visible gaps between numbers. Switch workspaces (MOD+N and MOD+J/K chains) — the underline must follow within one bar poll (0.15s), the occupancy colors must track window opens/closes/moves, and minimizing the only window on a workspace must mute it.
- **Bar desync soak (the v5 cap):** after a bar restart, run `grep -c Malformed ~/.cache/rhizome/bar.log` after 5 minutes of normal use — must read 0. The bar process binds wl_compositor at v5 (wm-protocol.lisp has the same cap) because wayflan cannot demarshal wl_surface v6 events; a single v6 event desyncs the bar's stream and kills it.
- **Pointer warp on switch:** MOD+2 — the pointer must jump to the output's center within the same frame as the workspace change. Disable it live (`(rhizome:configure '(:pointer (:warp-on-switch nil)))`) and confirm MOD+3 leaves the pointer where it was. Re-enable after.
- **configure_bounds (new windows):** open an app that tends to come up huge (an image viewer with a large image, some terminal profiles) — its initial size must stay within the screen even before rhizome proposes tiled dimensions.
- **Recording indicator:** if `:recording` is in a bar segment, start a screencast (portal-based screen recorder) — a red REC must appear within one bar poll and vanish when it stops.
- **Remembered placements (v4 identifiers):** open an app, move it to workspace 5, float it, close it — then reopen it: it must land back on workspace 5, floating, without any rule. Log out and back in with it open: same result. `cat ~/.local/share/rhizome/window-state.sexp` after closing shows the entry.
- **Presentation mode:** `(rhizome:configure '(:output (:presentation-mode :async)))` — the log must show the mode sent on the next render; a fast-scrolling page should now tear. Revert to `:vsync` and confirm the resend in the log.

- **xcursor theme:** set `(:pointer (:xcursor-theme "Adwaita" :xcursor-size 32))` before login — compositor-rendered cursors (resize edges, window borders) must follow; client-drawn cursors may not (per the protocol's own note).
- **WM hot-swap (detach):** open a few windows, note their workspaces, then `rhizome-cmd detach`. The bar must vanish, the WM process must exit ("detach confirmed" in `~/.cache/rhizome/rhizome.log`), and river must stay up with the windows frozen in place. Then take over again from a terminal: `sbcl --eval '(ql:quickload :rhizome)' --eval '(rhizome:run)' &` — the bar must come back and identified windows must land on their remembered workspaces. This is river's headline feature; if it works, everything above it works.
- **Workspace types:** `rhizome-cmd set-workspace-type floating 4` — every window on workspace 4 must float immediately; opening new windows there must float them too. `set-workspace-type tile 4` must return them all to the grid. Per-workspace layouts: MOD+Space to :even on workspace 1, switch to workspace 2 — it must still show its own layout; switch back — :even must be waiting. A restart of the WM must remember both (config-declared types via (:workspaces (:types ...)) and layouts per-workspace in-session).
- **Gaps:** tiled windows must show breathing room ((:gaps :inner 6), default): twice the gap between adjacent windows, once at screen edges, none around floats. Set (:gaps (:inner 0)) and confirm the classic flush tiling returns on the next manage pass.
- **Translucency:** the bar, launcher (MOD+D), message window (C-t m) and prompts (MOD+W) must render translucent ((:bar :opacity 0.9) and (:launcher :opacity 0.9), default) with correctly-weighted text — no washed-out colors (the buffers are premultiplied ARGB). Set (:bar (:opacity 1.0)) live and confirm surfaces go opaque without a restart (next redraw).
- **Launcher icons:** every entry must show its .desktop icon left of the text, vertically centered, with text aligned across iconless entries. (:launcher :icon-size 20) scales them.
- **Animations:** MOD+2 — the new workspace's windows must glide in from the screen edge (direction follows the switch), landing exactly in place within 200ms; no window may freeze mid-air (a missed frame snaps to the target). C-t Return (expose) — windows glide into the grid; pick one — it glides back. `rhizome-cmd maximize` glides to full-screen and back. Set `(:animations (:enabled nil))` live and confirm everything becomes instant (the old behavior); re-enable after.
- **Workspace-switch soak (the WM-death regression):** the exact scenario that terminated the WM twice: MOD+2, open a window there (Vivaldi), MOD+1 back — the WM must survive. Then rapid round-trips (2, 1, 3, 1 in fast succession, interrupting animations mid-flight) ×10; `grep -c 'invalid modification' ~/.cache/rhizome/rhizome.log` must not grow, and no fractional type errors (`is not of type (SIGNED-BYTE`) may appear. The failures were buffer-timing dependent: SPEED IS THE TEST.
- **MOD+Q closes cleanly:** close a window via MOD+Q and via `rhizome-cmd close` — the window must vanish within one manage cycle and the WM survive both paths.
- **Pointer window menus:** MOD+Middle-click on several windows (one tiled, one floating) — a context menu opens AT the pointer with Minimize, Float/Tile, Maximize, Fullscreen, Close, and Move-to-workspace entries. Hover must highlight a row (accent edge); click must run the action (verify Minimize hides and Restore returns it, Move-to-workspace relocates and the WM survives); Escape must dismiss; moving the pointer off the menu must dismiss it. Arrows + Enter select via keyboard. The close path routes through the manage-sequence queue — the WM must survive every action here.
- **C-t F11 fullscreens on the FIRST press:** the fullscreen request flush was once inverted (first press sent a no-op exit-fullscreen; only the second worked). Press C-t F11 once — fullscreen immediately; press again — back. Both directions on the first press, no double-press workaround.
- **Built-in screenshot (bar included):** `MOD+Print` / `rhizome-cmd screenshot` captures the full composited output INCLUDING the bar (grim copies the output

frame — layer surfaces are part of it). For a capture of an overlay state: `sleep 8 && rhizome-cmd screenshot` in a terminal, then open the target state (workspace
1. MOD+D launcher with a filter typed in) before the timer fires.

Flameshot on river is best-effort only: its editor floats via window rule; prefer the built-in path for anything important.

- **Pseudo-shadows on floats (river_decoration_v1):** MOD+drag a tiled window to float it — a soft shadow band must appear around it immediately (12px default, accent-tinted when focused: focus another window and the shadow's color shifts). Resize the float (MOD+Right-drag) — the shadow follows with no flicker. Un-float (menu → Tile, or `C-t f`) — the shadow must vanish. Close a floated window — the WM survives; `grep decoration ~/.cache/rhizome/rhizome.log` stays clean of errors. Tweak via `(:decorations (:radius 14 :alpha 0.5 :focused-alpha 0.7))`; `(:enabled nil)` makes floats flat again.
- **Tray right-click menus (dbusmenu):** right-click a tray icon on the bar (flameshot, variety) — the app's own context menu opens AT the icon (the pointer menu: hover highlights, click selects, Escape or pointer-off dismisses). Selecting an entry performs the app's action (e.g. flameshot's Capture / Open Launcher / Quit). Items with submenus appear as indented entries (a flat menu can't nest); separators and disabled entries are skipped.
- **Animation pump is quiet (performance):** switch workspaces a few times with apps open, run an expose — the glides must be smooth AND the WM log must stay nearly silent during them (the per-event [debug] lines are gated behind (:debug (:events t)); hundreds of lines per second of glide means the gate broke). During a sustained animation, CPU use stays modest (check with htop — the WM process should be well under a core).
- **WM hot-swap (the live upgrade):** `rhizome-cmd detach` — the WM serializes the session (placements, frame trees, per-output workspaces, focus), completes river's stop/finished handshake, and exits; the river-init supervision loop relaunches a fresh process  15s later. Expect:  10-20s of NO keybindings during the boot (river keeps rendering; the mouse works), then the desktop back EXACTLY as it was — same workspace showing, same windows in the same workspaces and frames, focus restored, bar back. The takeover file ( /.local/share/rhizome/takeover.sexp) is consumed by the restore. The same loop recovers a crashed WM — a WM death is a blip, not a hard reboot.

# 5 Appendix A: capture guide (how to report a failure)

- Section number + exact steps + what happened vs expected.
- `tail -50 ~/.cache/rhizome/rhizome.log` (WM) and `~/.cache/rhizome/bar.log`.
- For input/mode bugs: which modes were active (C-t h b lists minor modes), whether a timeout or manual exit was involved.
- One-liner repro if you can: the `rhizome-cmd` or key sequence.

# 6 Appendix B: quick reference

| Key | Action | Key | Action |
|---|---|---|---|
| MOD+Return | Terminal | C-t s / S | Split / stacked split |
| MOD+J / K | Cycle frame window | C-t R | Remove split |
| MOD+W / G / X | Windowlist / picker / palette | C-t +/- | Resize split |
| MOD+D | Launcher (fuzzy) | C-t r | Interactive resize |
| MOD+B | Toggle bar | C-t Q | Only (one frame) |
| MOD+1..9 | Workspace | C-t B | Balance splits |
| MOD+Shift+1..9 | Move window → ws | C-t Return | Expose |
| MOD+Shift+L | Lock | C-t f | Float/tile |

| | | | |
|---|---|---|---|
| MOD+Shift+E | Logout (×2) | C-t # | Mark window |
| C-t q | Quit (×2) | C-t m | Last message |
| C-t h | Help chains | C-t C | Command mode |

Text size: 10pt. Print me, tick me, break things.